

NGOC-AU DOAN

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SUMMARY

Embedded Software Engineer with 5 years of dedicated experience in software development and research. Hands-on experiment in **Camera Streaming (Deepstream/Holoscan)**, **Graphic Rendering (Wayland/Weston)**, and **Computer Vision**. Proven track record in complex domains including **Intelligent Transportation Systems (ITS)** and **IVI Automotive** and **5G Physical Layer**. Currently a GKS Scholar and Researcher at ISLAB, focused on Computer Vision, Deep Learning, and lightweight AI models for edge-device deployment.

SKILLS

Programming Languages	C/C++, Python, Bash Script
Computer Vision & AI	OpenCV, Deep Learning, Object Detection, AI Streaming
Graphics & Multimedia	Wayland/Weston, GPU Pipeline, Camera Streaming (DeepStream/Holoscan)
Hardware	PCB Design, MCU (STM32, PIC, ATmega), FPGA (Xilinx Virtex-7)
Domain Expertise	5G Physical Layer (3GPP), Intelligent Transportation Systems (ITS), Edge AI
Tools & Infrastructure	Docker, Kubernetes, CI/CD, Git, GTest/fff, CodeQL
Software Engineering	Data Structures, OOP, SOLID, Design Patterns, SDLC
Languages	English (TOEIC 800), Korean (TOPIK 3)

EXPERIENCE

Research Assistant Intelligent Systems Laboratory (ISLAB), University of Ulsan	Sep 2025 - Present <i>Ulsan, South Korea</i>
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- Conduct research in **Computer Vision**, **Deep Learning**, and **Edge AI** for intelligent perception systems.
- Develop and optimize real-time image processing and object detection algorithms for resource-constrained platforms.
- Design and implement high-performance **C++** vision pipelines for research and deployment.

Senior Embedded Software Engineer Hitachi Digital Services	Jul 2024 - Oct 2025 <i>Hanoi, Vietnam</i>
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- Implemented **HMAX** (Streaming and Analysis application) using **Holoscan SDK/Deepstream SDK** on Nvidia IGX/AGX Orin for Rail transport and Collaborated with Hitachi Rail and Nvidia teams.
- Develop tool applications for Set-top boxes.
- Analyzed and resolved **CodeQL** security issues for Windows printer drivers and implemented Unit Tests.

Embedded Software Engineer Eastgate Software Vietnam	May 2023 - Jul 2024 <i>Hanoi, Vietnam</i>
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- Developed and maintained **Intelligent Transportation Systems (ITS)** for Yunex Traffic, including application watchdogs and traffic actuation features.
- Built AI recognition systems capturing frames from **Basler/GigE Cameras** via Aravis and Deepstream SDK.
- Developed Windows simulation applications and deployed services using **Docker** containers.

Embedded Software Engineer Bosch Global Software Technologies	Sep 2022 - Apr 2023 <i>Hanoi, Vietnam</i>
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- Developed Video-in applications from **RTSP streams** using FFmpeg on Jetson Xavier NX.
- Maintained and up-streamed versions of **Wayland-ivi-extension**; implemented unit tests using **gtest** and **fff**.
- Gained deep expertise in Wayland protocol, Weston, and Graphics rendering pipelines.

Embedded Software Engineer
HCLTech

Jun 2021 - Sep 2022
Hanoi, Vietnam

- Developed **SS Block** (SSS, PSS, PBCH) in the **5G Physical Layer** based on 3GPP specifications.
- **Led a team of 5** to develop the **RRC connection** establishment procedure using ASN.1 and UDP protocol.
- Analyzed **free5GC** open source to support 5G core implementation and testing.

EDUCATION

University of Ulsan, Ulsan, South Korea

Sep. 2025 – June 2028

M.Sc. in Electrical, Electronic and Computer Engineering

- Awarded the **Global Korea Scholarship (GKS)**, covering 1 year of Korean language training and 2 years of graduate study.

Vietnam Maritime University, Haiphong, Vietnam

Sep. 2017 – June 2021

B.Sc. in Electronics and Telecommunications Engineering

- **Classification:** Excellent — **GPA:** 3.77/4.00
- **Scholarship:** Merit-Based Scholarship for the Top 6% of Outstanding Academic Students over 4 years.

HONORS AND AWARDS

- **Korea Government Scholarship**, National Institute for International Education 2025
- **Certificate of Merit for Excellent Student**, Vietnam Maritime University 2017 – 2021
- **Third Prize**, Math Olympic at Vietnam Maritime University level 2021
- **Second Prize**, Scientific Research Students at Vietnam Maritime University level 2020
- **Second Prize**, Math Olympic at Vietnam Maritime University level 2019
- **Entrance Valedictorian**, Electric and Electronic Engineering, Vietnam Maritime University 2017

PUBLICATIONS

- Nguyen Manh Cuong, **Doan Ngoc Au**, Quang Nhat Doan, Phuc Hong Bui, Danh Cong Do, Hung Pham-Viet, [“A Real-time Drowsiness Detection System Based on Raspberry”](#), poster paper in 2020 IEEE eighth International Conference on Communications and Electronics (ICCE), 2020.